

Taehyeon Kim

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PROFILE

Third-year Computer Science specialist at the University of Toronto (GPA 3.65/4.00), focused on machine learning and system architecture. Builds full-stack systems end to end, from production backend infrastructure and billing systems to ML pipelines, models, and fraud-detection experimentation, with deliberate attention to tradeoffs rather than just working code.

EDUCATION

University of Toronto

2024 – Present

B.Sc. Computer Science Specialist · GPA: 3.65 / 4.00

TECHNICAL SKILLS

Languages: Python, Java, C, TypeScript, Rust, Bash, SQL

Frameworks & Tools: FastAPI, Next.js, React, Axum/Tokio, PostgreSQL, Git, Linux, SQLAlchemy, Auth0, Tailwind CSS, Supabase

Cloud: AWS (IAM, VPC, EC2, S3, Lambda); pursuing AWS Certified Solutions Architect Associate

Concepts: System Design, OOP, Design Patterns, Data Structures & Algorithms, Scrum / Agile, REST APIs

AI / ML: NumPy, scikit-learn, Gemini API, LLM integration, anomaly detection (Isolation Forest, Benford's Law, BiLSTM)

PROJECTS & HACKATHONS

MatiMath (AI Math Tutoring App)

May 2026 – Present

Collaborative project · Team of 3

- Leading technical development of a Flutter tablet app that pairs an AI tutor character with step-based math exercises (fill-in-blank, matching, drawing, handwriting) and real-time TTS narration.
- Built Google Play Billing with server-side receipt verification through Firebase Cloud Functions as the sole writer of purchase entitlement, and cut the baked TTS asset pipeline from 476MB to 353MB.
- Wrote the privacy policy and account-deletion flow required for Google Play Families compliance.

chalk-talk (AI Blackboard Tutor Research)

2026 – Present

Solo research project

- Designing a closed-loop tutoring architecture where a vision-language model reads the canvas each step and decides the next spoken phrase or stroke, instead of generating a full script upfront.
- Built a working MVP that pre-generates a full lesson as JSON and animates it stroke-by-stroke with synced narration, used as the baseline and training-data source for the closed-loop research.

UTM Degree Explorer (Local-First Degree Planner)

May 2026

Solo project

- Built a full-stack degree planner over a fully scraped UTM Academic Calendar (2,346 courses, 164 programs), with a rule-based parser converting prerequisite text into structured ASTs at 86% coverage.
- Implemented live prerequisite validation and a React/Zustand planner UI with Supabase-backed multi-plan sync and guest mode.

Yosemite (AI Compliance & Fraud Detection Platform)

March 2026

GenAI Genesis 2026 · Team of 4

- Built the Rust (Axum) fraud-report API and integrated Google Gemini Vision and an LLM service for automated invoice and document analysis.
- Built the Next.js risk dashboard (risk overview, flagged transactions, protection score) and the PDF upload flow for the compliance platform.

Multi-Class Painting Classification Model (CSC311 Course Project)

Winter 2026

University of Toronto · Team of 4

- Ran feature-engineering ablations (StandardScaler normalization, TF-IDF vs. numeric-only) across three model families on a 3-class painting-classification task, showing TF-IDF removal dropped Bernoulli Naive Bayes accuracy from 93.5% to 76.7% and shaping the final unscaled, full-feature preprocessing pipeline.
- Analyzed and wrote up feature distributions (numeric, Likert, categorical) across 1,686 survey responses; the team's final neural network reached 94% test accuracy (macro F1 0.94).

Tango (Reinvented Job Matching Platform)

February 2026

DeerHacks V 2026 · Team of 3

- Implemented the core swipe/match feature (role-based discover queue, swipe UI, recruiter inbox) and integrated Auth0 authentication on a Next.js 15 frontend.
- Built a real-time chat feature with a matches sidebar and unread-message badges, and migrated image uploads from filesystem storage to PostgreSQL.

WORK EXPERIENCE

Crew Member

June 2026 – September 2026

Chipotle Mexican Grill

- Ran a high-volume, fast-paced kitchen line, building strong time management and prioritization skills while delivering friendly, consistent customer service during peak rush periods.

Math, Physics & Chemistry Tutor

September 2024 – 2026

Self-employed

- Provided personalized academic support to high school students; adapted explanations to individual learning gaps rather than following fixed curricula.

Customer Service Associate

Summer 2025

Korean Supermarket

- Developed conflict resolution and negotiation skills in a high-volume customer-facing environment.

LEADERSHIP

Finance Associate

2026 – Present

UTM Mathematical & Computational Sciences Society (MCSS)

- Volunteer on the student executive team representing all full-time Mathematical & Computational Sciences majors at UTM, supporting audit planning and financial reporting.
- Help run department events and day-to-day club operations alongside the team, in a volunteer capacity.

VP Finance

2025 – 2026

UTM Salsa & Bachata Club

- Managed the club's annual budget and financial operations for a year, overseeing event expenses, sponsorships, and financial reporting.

ADDITIONAL

Languages: Fluent in English and Korean**Presentations:** Weekly presentation experience over three years; comfortable communicating technical concepts to varied audiences.